

Battery & Resistor Circuits (completed)

Goal.

- Pin down what current and ΔV actually mean, and the four rules that follow just from charge and energy conservation.
- Turn Ohm's law into a tool: given a resistor and a ΔV across it, get I .
- Work a single resistor both with an ideal battery and with a real one (internal resistance r), to see what r costs you.
- Derive $R_S = R_1 + R_2$ for series and $R_P = (1/R_1 + 1/R_2)^{-1}$ for parallel from the rules in §1 — not as formulas to memorize, but as consequences of “same current” and “same ΔV .”

1. Definitions

Current: the amount of charge passing a point (through an element) per second. $I \equiv \Delta q / \Delta t$.

units: $1 \text{ A} = 1 \text{ C/s}$

Change in voltage, ΔV : the potential energy per charge gained or lost by a charge that crosses an element.

Series: elements connected back-to-back — one shared wire between them, so the same charge has to pass through each in turn.

Parallel: elements connected side-by-side, sharing *both* endpoints (the same two nodes).

2. Four rules

- Elements in series have the same current passing through them.
- Elements in parallel have the same voltage change (ΔV) across them.
- Current is conserved at a junction.
- Around any closed loop, the ΔV 's sum to zero.

Check. The last two rules are each a conservation law in disguise. Which one is charge conservation, and which one is energy conservation?

The junction rule (current conserved) is charge conservation — charge can't pile up or vanish at a junction, so whatever flows in must flow out. The loop rule (ΔV 's sum to zero) is energy conservation — walk all the way around back to where you started, and the net PE per charge you gained and lost along the way has to cancel.

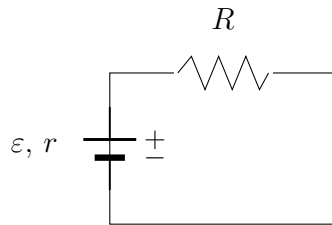
3. Ohm's law defines a resistor

$$\Delta V = IR$$

Given a resistor of resistance R and a voltage difference ΔV across it, the current through it is

$$I = \frac{\Delta V}{R}$$

4. One resistor and a battery



(a) **Ideal battery:** $\varepsilon = 12 \text{ V}$, $r = 0$, $R = 4 \text{ } \Omega$.

$$I = \frac{\varepsilon}{R} = \frac{12}{4} = 3.0 \text{ A} \quad V_B = \varepsilon - Ir = 12 - 0 = 12 \text{ V}$$

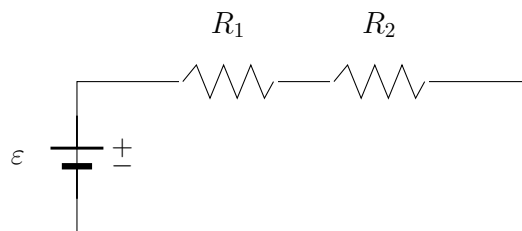
(b) **Real battery:** same ε and R , but now $r = 1 \text{ } \Omega$.

$$I = \frac{\varepsilon}{R + r} = \frac{12}{5} = 2.4 \text{ A} \quad V_B = \varepsilon - Ir = 12 - (2.4)(1) = 9.6 \text{ V}$$

Check. V_B dropped in part (b). Where did that energy per charge go?

Into the battery's own internal resistance r — the missing $12 - 9.6 = 2.4 \text{ V}$ is exactly $Ir = (2.4 \text{ A})(1 \text{ } \Omega) = 2.4 \text{ V}$. That energy per charge is dissipated *inside* the battery itself, not delivered to R .

5. Two resistors in series



Derive R_S (symbols only — use the series current rule and Ohm's law on each resistor, then the loop rule):

Series rule: the same current I flows through both resistors. Ohm's law on each: $\Delta V_1 = IR_1$, $\Delta V_2 = IR_2$. Loop rule:

$$\varepsilon = \Delta V_1 + \Delta V_2 = IR_1 + IR_2 = I(R_1 + R_2).$$

But by definition of the single equivalent resistor, $\varepsilon = IR_S$. Comparing the two:

$$\boxed{R_S = R_1 + R_2}$$

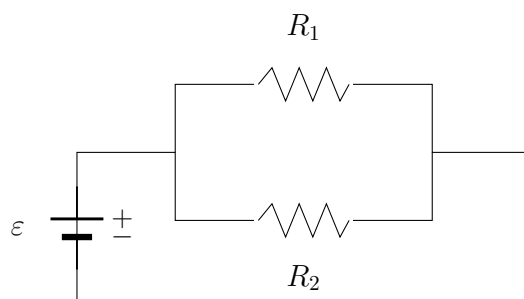
Now with numbers: $\varepsilon = 12 \text{ V}$, $R_1 = 4 \text{ } \Omega$, $R_2 = 8 \text{ } \Omega$ (ideal battery).

$$\begin{aligned} R_S &= 4 + 8 = 12 \text{ } \Omega & I &= \frac{\varepsilon}{R_S} = \frac{12}{12} = 1.0 \text{ A} \\ \Delta V_1 &= IR_1 = 4.0 \text{ V} & \Delta V_2 &= IR_2 = 8.0 \text{ V} \end{aligned}$$

Check. Check the loop rule with your two ΔV 's above: do they add up to ε ?

Yes: $4.0 + 8.0 = 12 \text{ V} = \varepsilon$. That's not a coincidence — it's exactly the loop rule, built into how R_S was derived.

6. Two resistors in parallel



Derive R_P (symbols only — use the parallel voltage rule and Ohm's law on each resistor, then the junction rule at the top node):

Parallel rule: both resistors see the same ΔV (here, $\Delta V = \varepsilon$, since each branch connects straight across the ideal battery). Ohm's law on each: $I_1 = \Delta V/R_1$, $I_2 = \Delta V/R_2$. Junction rule at the top node:

$$I = I_1 + I_2 = \frac{\Delta V}{R_1} + \frac{\Delta V}{R_2} = \Delta V \left(\frac{1}{R_1} + \frac{1}{R_2} \right).$$

But by definition of the single equivalent resistor, $I = \Delta V/R_P$. Comparing the two:

$$\boxed{R_P = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}}$$

Now with numbers: $\varepsilon = 12 \text{ V}$, $R_1 = 4 \text{ } \Omega$, $R_2 = 12 \text{ } \Omega$ (ideal battery).

$$\begin{aligned} R_P &= \left(\frac{1}{4} + \frac{1}{12} \right)^{-1} = \left(\frac{3}{12} + \frac{1}{12} \right)^{-1} = \left(\frac{4}{12} \right)^{-1} = 3.0 \text{ } \Omega \\ I_1 &= \frac{\varepsilon}{R_1} = \frac{12}{4} = 3.0 \text{ A} & I_2 &= \frac{\varepsilon}{R_2} = \frac{12}{12} = 1.0 \text{ A} \\ I_{\text{total}} &= I_1 + I_2 = 4.0 \text{ A} \quad (\text{check: } \varepsilon/R_P = 12/3 = 4.0 \text{ A } \checkmark) \end{aligned}$$

Check. Which resistor carries more current, and why does that make sense given which one is smaller?

R_1 ($4\ \Omega$) carries more current (3.0 A vs. 1.0 A). Both resistors feel the *same* ΔV , and $I = \Delta V/R$ — so the smaller resistance lets more current through for that same push. Smaller R means less obstruction, not less current.